Resilience Academy

SDI Training for ICT Professionals

QGIS & CRD Integration

Your Desktop Window into the Climate Risk Database

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Day 4 | 1 Hour Lecture | crd.resilienceacademy.ac.tz

Resilience Academy

Session Overview

What We Will Cover

- What is QGIS and why it matters for SDI
- How QGIS connects to CRD (two ways: WMS and WFS)
- Step-by-step: adding CRD layers to QGIS
- Loading raster imagery from CRD
- Combining multiple layers for analysis
- Working with CRD data even without internet

Key Idea

QGIS is your desktop window into CRD.

Everything you uploaded to CRD during Days 2–3 can be viewed, analysed, and turned into professional maps using QGIS — a free application on your computer.

No Programming Required

Everything today uses **menus, buttons, and clicks**. No code, no command line.

What is QGIS?

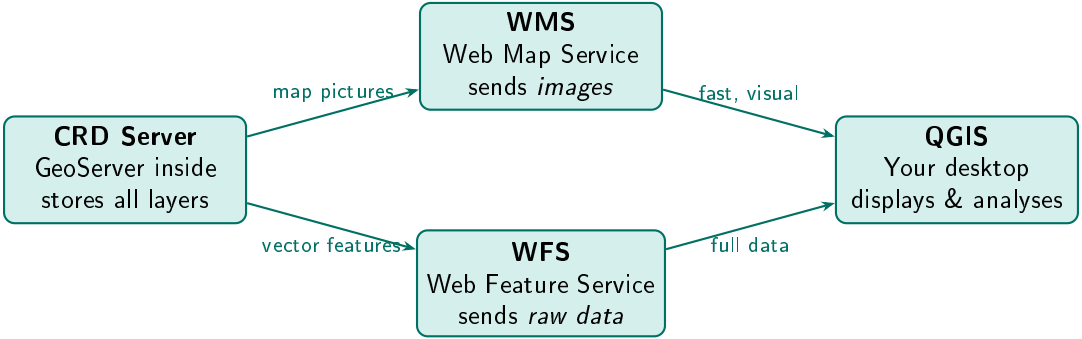
QGIS in Simple Terms

- **Free and open-source** Geographic Information System
- Think of it as **Google Earth Pro, but much more powerful**
- You can open maps, analyse data, create reports, and print professional maps
- Runs on Windows, Mac, and Linux
- Used by governments, universities, and NGOs worldwide

What Can You Do with QGIS?

1. **View** — Open maps, satellite images, survey data
2. **Query** — Ask questions: "Which wards have more than 5 schools?"
3. **Style** — Change colours, symbols, and labels
4. **Analyse** — Measure distances, count features, find patterns
5. **Print** — Create PDF maps with titles, legends, and scale bars

How QGIS Connects to CRD



WMS (Map Images)

- CRD sends a **picture** of the map
- Fast to load, looks nice

WFS (Raw Vector Data)

- CRD sends the **actual data points, lines, polygons**

Step-by-Step: Adding a WMS Connection

Follow These Steps in QGIS

1. Open QGIS and start a new project (**Project** → **New**)
2. Go to **Layer** → **Add Layer** → **Add WMS/WMTS Layer...**
3. Click the **New** button to create a new connection
4. Fill in the details:
   - **Name:** `CRD GeoServer`
   - **URL:** `https://crd.resilienceacademy.ac.tz/geoserver/ows`
5. Click **OK**, then click **Connect**
6. Wait for the layer list to load (this may take a few seconds)
7. Browse the list — you will see all layers published on CRD
8. Select a layer (e.g., `Tanga_imagery_reduced`) and click **Add**

Step-by-Step: Adding a WFS Connection

Follow These Steps in QGIS

1. Go to **Layer** → **Add Layer** → **Add WFS Layer...**
2. Click **New** to create a new connection
3. Fill in the same details:
   - **Name:** `CRD GeoServer WFS`
   - **URL:** `https://crd.resilienceacademy.ac.tz/geoserver/ows`
4. Click **OK**, then click **Connect**
5. From the layer list, select:
   - `mwanza_businesses`
   - `mwanza_education_facilities`
6. Click **Add** — QGIS will download the data

What You Get

WMS vs WFS: Side-by-Side Comparison

WMS (Map Images)

- ✓ Fast loading speed
- ✓ Good for basemaps
- ✓ Low bandwidth usage
- × Cannot click features
- × Cannot open attribute table
- × Cannot export data
- × Cannot change styling

WFS (Raw Data)

- ✓ Full vector data
- ✓ Click & identify features
- ✓ Open attribute table
- ✓ Export to file
- ✓ Change colours & symbols
- × Slower to load
- × Uses more bandwidth

Rule of Thumb

Use **WMS** for background layers you just want to see. Use **WFS** for layers you want to **work**

Loading Raster Imagery from CRD

What is Raster Data?

- Raster = a grid of pixels, like a photograph
- Satellite images, aerial photos, elevation maps
- On CRD, rasters are served through **WMS**

Load Tanga Imagery in QGIS

1. Use the WMS connection you already created
2. **Layer** → **Add WMS/WMTS Layer**
3. Select `CRD GeoServer` and click **Connect**
4. Find `Tanga_imagery_reduced` in the list
5. Click **Add**

Why Use Raster from CRD?

- You get **up-to-date imagery** without downloading large files
- Perfect as a **background** for your vector layers
- Zoom in to see detail — CRD streams tiles on demand

Tip

If the raster looks blurry when zoomed in, right-click the layer → **Properties** → **Rendering** → set tile size to 512 or

Loading Your Local CRD Layers in QGIS

Remember Day 2?

On Day 2, you installed CRD on your own computer. That local CRD instance also has a GeoServer built in. You can connect QGIS to **your own local CRD** the same way.

Steps for Local Connection

1. Make sure your local CRD is running (`http://localhost` in your browser)
2. In QGIS: **Layer** → **Add WFS Layer**
3. Click **New**, enter:
   - **Name:** `My Local CRD`
   - **URL:** `http://localhost/geoserver/ows`
4. Click **Connect**, and [illegible]

Why Connect Locally?

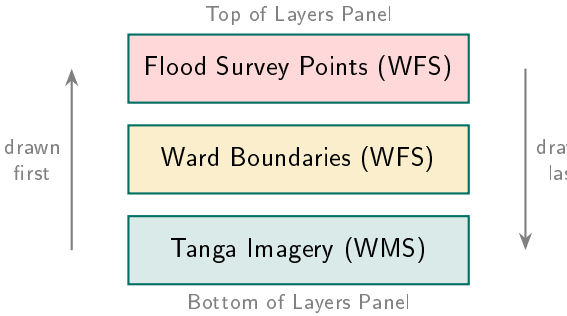
- **No internet needed** — everything is on your machine
- **Faster** — data does not travel across the network
- **Private** — your data stays on your computer
- Perfect for testing before publishing to

Combining Layers: Building a Complete Map

Layer Stacking Example

Build this map in QGIS by adding layers in order:

1. **Bottom:** Tanga satellite imagery (WMS)
   Provides the visual context
2. **Middle:** Ward boundaries (WFS)
   Shows administrative areas as outlines
3. **Top:** Flood survey points (WFS)
   Shows where data was collected



How to Reorder Layers

In the **Layers Panel** (left side), simply **drag**

Result

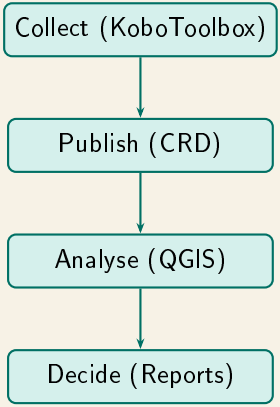
You see flood locations on top of ward

Why QGIS + CRD Integration Matters

What You Can Do Now

1. **Analyse CRD data offline**
   Download via WFS, work without internet
2. **Create professional reports**
   Print layouts with legends, titles, logos
3. **Combine data sources**
   Mix CRD layers with local files, GPS tracks, Excel tables
4. **Answer complex questions**
   "How many schools are within 500m of a flood zone?"
5. **Share results**
   Export to map or PDF, ...

The Bigger Picture



Session Summary

What We Learned

- QGIS is a free, powerful desktop GIS that connects directly to CRD
- **WMS** gives you map images — fast and visual, ideal for basemaps
- **WFS** gives you raw vector data — you can query, style, and export
- Both use the same GeoServer URL: `.../geoserver/ows`
- You can connect to the **online CRD** or your **local CRD** (same steps, different URL)
- Combining raster backgrounds with vector data creates **complete analytical maps**

Key URLs to Remember

- **Online:** `https://crd.resilienceacademy.ac.tz/geoserver/ows`
- **Local:** `http://localhost/`

Next Up

Session 2: Map Making in QGIS

We will style these layers, add labels, and create a professional print-ready map.